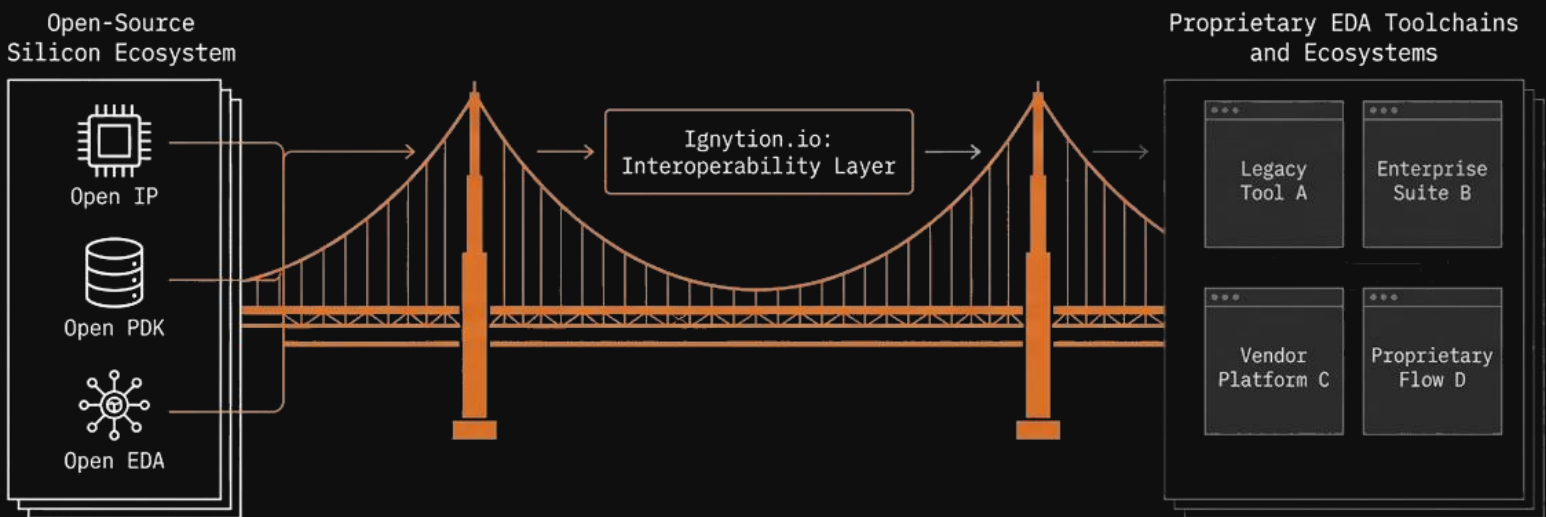




Ignnytion's Guide to Open Source

A platform thesis on infrastructure, orchestration, and the execution layer for next-generation silicon development.

What's inside.

01	Introduction	p. 03
02	The Compute Layer	p. 04
03	Orchestration & Execution	p. 06
04	The Workflow Surface	p. 08
05	Observability & Trust	p. 10
06	Looking Ahead	p. 12
07	About Ignition	p. 13

Silicon development needs a **platform**, not another tool.

Modern silicon teams are stitching together compute, orchestration, and verification across an expanding surface of vendors, formats, and environments. Ignition is the platform layer beneath that work.

This whitepaper outlines the principles behind the Ignition platform — what we call the *ignition layer* — and the architecture that makes it possible. We cover the four primitives that define the platform: compute, orchestration, workflow surface, and observability.

Each section is structured the same way: a short framing of the problem, followed by the specific capabilities Ignition exposes and how teams typically adopt them.

"The interface remains quiet. The system speaks through clarity. The orange appears only when something comes alive."

Ignition is built for engineering organizations who treat their infrastructure as a first-class product surface — and who expect their platform to behave the same way.

The Compute Layer.

The compute layer is the foundation of any silicon platform — a deterministic substrate that exposes capacity, isolation, and cost as first-class primitives. Three building blocks define how Ignition approaches it.

Primitives of the compute layer.

02 · BUILDING BLOCKS

Capacity Fabric

WHAT IT IS

A pooled, multi-tenant capacity substrate that exposes CPU, GPU, and accelerator resources as a single addressable surface, regardless of underlying cloud or on-prem deployment.

WHAT IT'S USED FOR

Sizing, scheduling, and rebalancing simulation, synthesis, and verification workloads against shared capacity — without vendor-specific job descriptions.

WHY IT MATTERS

Teams stop building bespoke schedulers per environment. The fabric makes capacity tradeoffs explicit and auditable.

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ignition.io/platform/compute

Deterministic Runtimes

WHAT IT IS

Immutable, content-addressed runtime images that pin every toolchain, library, and license configuration for a given workload.

WHAT IT'S USED FOR

Reproducing engineering environments precisely — across machines, across teams, and across years. Regulatory and tape-out review workflows rely on this guarantee.

WHY IT MATTERS

"It worked on my machine" becomes a structural impossibility, not a cultural problem.

LEARN MORE

ignition.io/platform/runtimes

Orchestration & Execution.

Orchestration is the moment a system comes alive. Ignition treats it as a first-class signal — the place where intent becomes activity, where the orange appears.

How execution flows through the platform.

03 • EXECUTION MODEL

Intent Graphs

WHAT IT IS

A declarative, typed representation of a workload — what should happen, against what inputs, under which constraints — without specifying how.

WHAT IT'S USED FOR

Describing build-and-verify pipelines, regression sweeps, and physical-design flows in a portable way. Intent graphs are versioned alongside RTL.

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ignition.io/platform/intent

Activation Pipelines

WHAT IT IS

The execution engine that resolves an intent graph against available capacity, lineage, and license posture — and produces a deterministic activation plan.

WHAT IT'S USED FOR

Running long-horizon flows that need to survive partial failure, re-entry, and cross-team handoff without losing state.

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ignition.io/platform/activation

The Workflow Surface.

The surface where humans and the platform meet. Calm by default — activated only at moments of decision, review, or handoff.

How teams work inside Ignition.

The Control Console

WHAT IT IS

A single, low-noise console that exposes capacity, activations, and lineage at the granularity teams actually work at.

WHY IT MATTERS

Operators stop swivel-chairing between dashboards. Engineering leads see one truthful state.

LEARN MORE

ignition.io/platform/console

Review & Handoff

WHAT IT IS

First-class review surfaces — for sign-off, regression diffs, and tape-out gates — that preserve provenance end-to-end.

WHY IT MATTERS

Decisions made in the platform stay attached to the artifacts they were made about. Audits become a query, not a project.

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Observability & Trust.

Every activation, artifact, and decision is anchored to a verifiable lineage. Trust is not a feature on the platform — it is the surface on which the platform is built.

Provenance is the product.

Lineage Spine

WHAT IT IS

A content-addressed event log that records every activation, artifact, and decision across the platform.

WHAT IT'S USED FOR

Answering "what produced this?", "who approved it?", and "what would change if we re-ran from here?" with mathematical certainty.

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ignition.io/platform/lineage

Signal Layer

WHAT IT IS

The observability primitives — metrics, traces, and structured events — exposed as a typed surface, not a free-text log stream.

WHAT IT'S USED FOR

Building diagnostic, capacity, and SLO views that hold up across months and across teams.

LEARN MORE

ignition.io/platform/signals

Where the **ignition layer** is going next.

The next phase of the platform is about quieter interfaces and louder guarantees — fewer dashboards, deeper provenance, and tighter coupling between intent and execution.

Three directions shape our roadmap: (1) richer intent graphs that encode physical-design constraints natively; (2) provenance primitives that extend across third-party vendor tools without wrapping them; and (3) a control console designed to disappear into the work.

We will publish detailed specifications for each as they reach the platform. This whitepaper is a snapshot of the principles. The principles are stable; the surface area continues to grow.

Black for depth. White for clarity. Orange for ignition.



The interface remains quiet. The system speaks through clarity.

Ignition is a platform for silicon development — engineered, stable, precise, infrastructure-grade. Talk to our team about deploying the ignition layer inside your organization.

TALK TO US —

[IGNITION.IO](https://ignition.io)